

Chapter 10 Gases

Chapter 11 Liquids and Intermolecular Forces

1. Which of the following statements about gases is false? E

A) Gases are highly compressible. ✓

B) Distances between molecules of gas are very large compared to bond distances within molecules. ✓

C) Non-reacting gas mixtures are homogeneous. ✓

D) Gases expand spontaneously to fill the container they are placed in. ✓

E) All gases are colorless and odorless at room temperature. ✗

2. Which of the following equations shows an incorrect relationship between pressures given in terms of different units? D

A) $1.20 \text{ atm} = 122 \text{ kPa}$ ✓

B) $152 \text{ mm Hg} = 2.03 \times 10^4 \text{ Pa}$ ✓

C) $0.760 \text{ atm} = 578 \text{ mm Hg}$ ✓

D) $1.0 \text{ torr} = 2.00 \text{ mm Hg}$ ✗

E) $1.00 \text{ atm} = 760 \text{ torr}$ ✓

$$1 \text{ atm} = 760 \text{ mmHg} = 760 \text{ torr} = 1.10325 \times 10^5 \text{ Pa} = 101.325 \text{ kPa} = 1.01325 \text{ bar}$$

3. Sodium bicarbonate is reacted with concentrated hydrochloric acid at 37.0 °C and 1.00 atm. The reaction of 6.00 kg of bicarbonate with excess hydrochloric acid under these conditions will produce E L of CO₂.

A) 1.09×10^2

B) 2.85×10^4

C) 1.82×10^4

D) 8.70×10^2

E) 1.82×10^3

4. The kinetic-molecular theory predicts that pressure rises as the temperature of a gas increases because E.

- A) the average kinetic energy of the gas molecules decreases
- B) the gas molecules collide more frequently with the wall
- C) the gas molecules collide less frequently with the wall
- D) the gas molecules collide more energetically with the wall
- E) both the gas molecules collide more frequently with the wall and the gas molecules collide more energetically with the wall

5. Arrange the following gases in order of increasing average molecular speed at 25 °C. C

He, O₂, CO₂, N₂

A) He < N₂ < O₂ < CO₂

B) He < O₂ < N₂ < CO₂

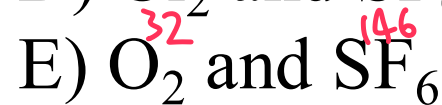
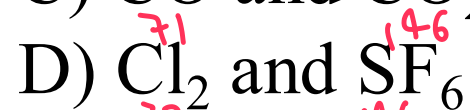
C) CO₂ < O₂ < N₂ < He

$m \uparrow \quad v \downarrow$

D) CO₂ < N₂ < O₂ < He

E) CO₂ < He < N₂ < O₂

6. A mixture of two gases was allowed to effuse from a container. One of the gases escaped from the container 1.43 times as fast as the other one. The two gases could have been D.



$$v \propto \sqrt{\frac{1}{M}}$$

7. A real gas will behave most like an ideal gas under conditions of **B** .

- A) high temperature and high pressure
- B) high temperature and low pressure**
- C) low temperature and high pressure
- D) low temperature and low pressure
- E) STP

8. In liquids, the attractive intermolecular forces are E.

A) very weak compared with kinetic energies of the molecules

B) strong enough to hold molecules relatively close together

C) strong enough to keep the molecules confined to vibrating about their fixed lattice points

D) not strong enough to keep molecules from moving past each other

E) strong enough to hold molecules relatively close together but not strong enough to keep molecules from moving past each other

9. The strongest interparticle attractions exist between particles of a _____, and the weakest interparticle attractions exist between particles of a **B** .

A) solid, liquid

B) solid, gas

C) liquid, gas

D) liquid, solid

E) gas, solid

10. Of the following substances, only D has London dispersion forces as the only intermolecular force.

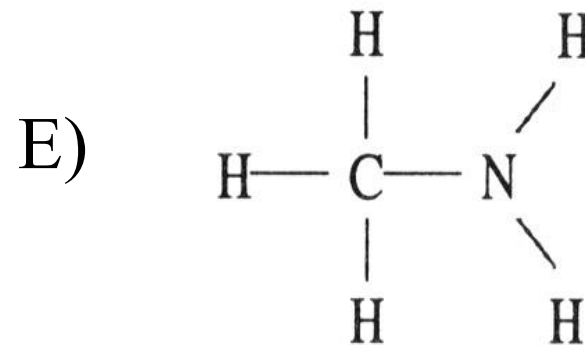
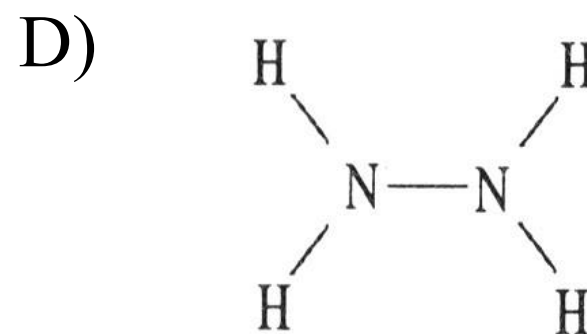
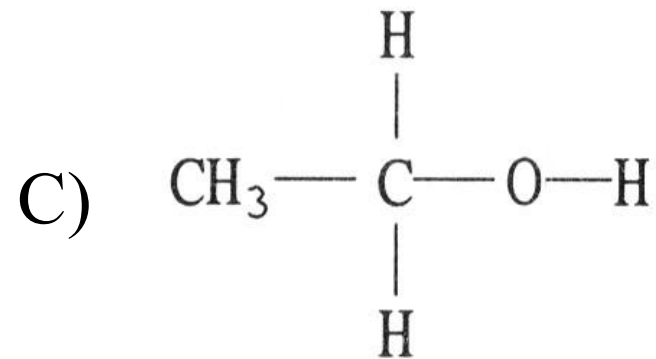
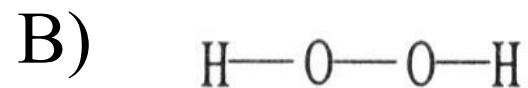
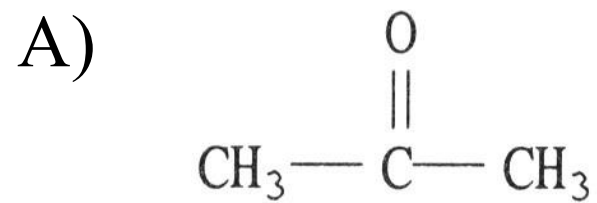
- A) CH_3OH
- B) NH_3
- C) H_2S
- D) Kr**
- E) HCl

} H-bond

Dipole

Lon

11. Which one of the following substances will not have hydrogen bonding as one of its intermolecular forces? A



12. Which statements about viscosity are true? E

- (i) Viscosity increases as temperature decreases. ✓
- (ii) Viscosity increases as molecular weight increases. ✓
- (iii) Viscosity increases as intermolecular forces increase. ✓

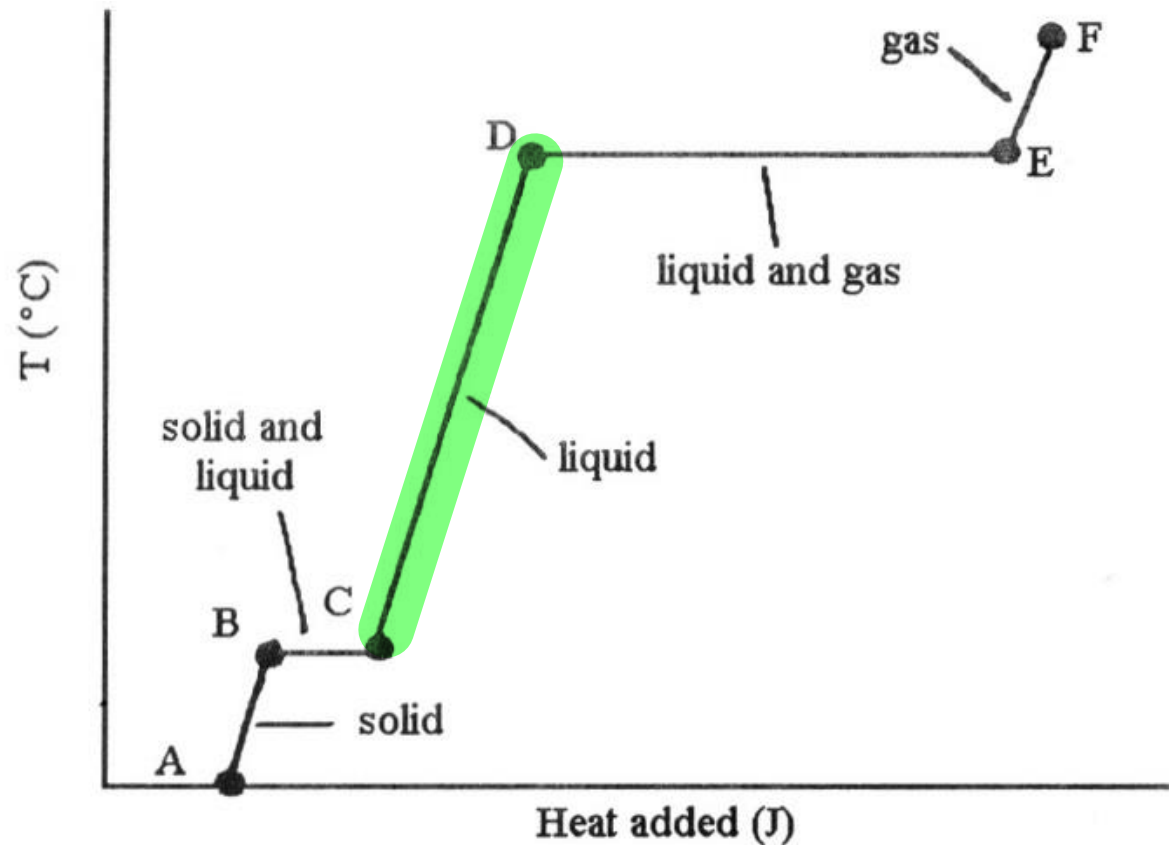
- A) (i) only
- B) (ii) and (iii)
- C) (i) and (iii)
- D) none
- ☒ E) all

13. How high a liquid will rise up a narrow tube as a result of capillary action depends on A.

- A) the magnitudes of cohesive forces in the liquid and adhesive forces between the liquid and the tube, and gravity
- B) gravity alone
- C) only the magnitude of adhesive forces between the liquid and the tube
- D) the viscosity of the liquid
- E) only the magnitude of cohesive forces in the liquid

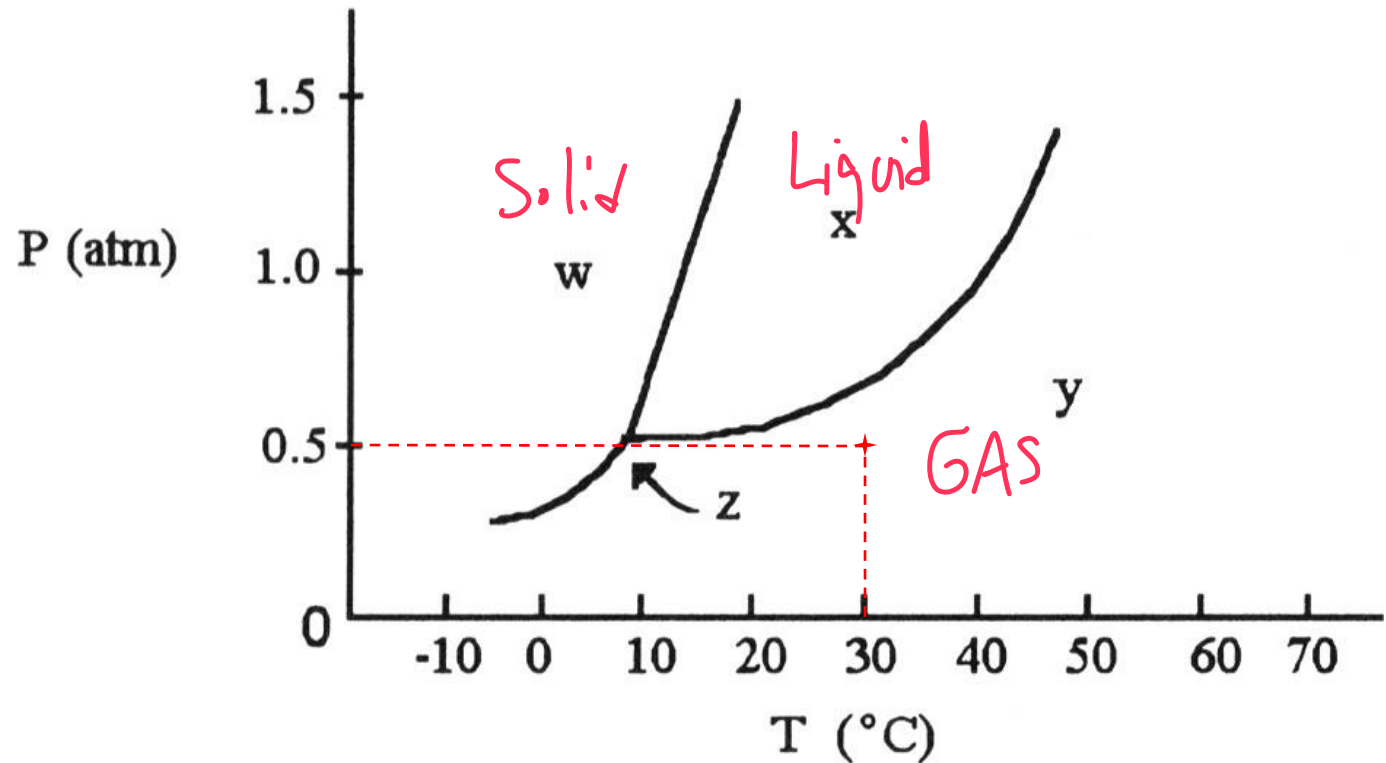
14. The heating curve shown was generated by measuring the heat flow and temperature for a solid as it was heated. The slope of the C segment corresponds to the heat capacity of the liquid of the substance.

- A) AB
- B) BC
- C) CD**
- D) DE
- E) EF



15. The phase diagram of a substance is given. This substance is a B at 30 °C and 0.5 atm.

- A) liquid
- B) gas**
- C) solid
- D) supercritical fluid
- E) crystal



16. The heat of fusion of water is 6.01 kJ/mol. The heat capacity of liquid water is 75.3 J/mol·K. The conversion of 50.0 g of ice at 0.00 °C to liquid water at 0.00°C requires B kJ of heat.

A) 6.01

☒ B) 16.7

C) 75.3

D) 17.2

E) Insufficient data are given.